



Hazards, Safety, & Risk Control

Area	Code / Standard	Requirement	Compliance Official	Governing Body
Fire/Explosion Protection	NFPA 85 / ANSI Z21.13	Safety valves, pressure relief, gas train regulators, and explosion-proof enclosures are required.	Licensed Burner Tech / Fire Marshal	NFPA, ANSI
Worker Safety & Hot Work	OSHA 1910 Subparts M, Q, S	PPE, Lockout/Tagout, confined space entry, and hot work permits must be enforced.	EHS Manager / OSHA Inspector	OSHA
Combustion Safety	CSD-1 (Controls & Safety Devices Code)	Required for gas/oil boilers under 12.5 million BTU/hr in many jurisdictions.	Combustion Engineer	ASME, State Agencies
Emergency Shutoff Systems	UL 353 / FM Approved	Must include tested emergency shutoff switches and interlocked combustion airflow switches.	Licensed Electrician / Boiler Inspector	UL, FM Global

Boiler Fabrication & Construction

Area	Code / Standard	Requirement	Compliance Official	Governing Body
Pressure Vessel Fabrication	ASME BPVC Section VIII	Must be constructed by an ASME-certified shop with U, S, or H stamp.	AI (Authorized Inspector)	ASME
Welding/Brazing Procedure Qualification	ASME Section IX	WPS and PQR documentation must be approved before production welding.	QA Engineer + Inspector	ASME
Post-Weld Heat Treatment	NBIC / ASME	Required on materials like P91, P22, or carbon steel > 1.5 in thick. Must follow time/temp curves.	Metallurgist / AI	NBIC
Shell Thickness / Head Design	API 650 / ASME BPVC	Must meet shell stress formulas and dish head stress radius per pressure class.	Design Engineer	ASME, API